MATH 110 — Week 6

Homework 6 — Solutions

HW6 P1.

Idea. The equation $P^2 = P$ controls eigenvalues and supplies a decomposition for every vector.

1. Assume the intended linearity hypothesis. If $Pv = \lambda v$ with $v \neq 0$, then $P^2 v = \lambda^2 v$, while $P^2 v = Pv = \lambda v$. Hence $\lambda(\lambda - 1) = 0$, so $\lambda \in \{0, 1\}$. It is not asserted that both occur.
2. Every $v = (v - Pv) + Pv$, with $P(v - Pv) = Pv - P^2 v = 0$ and $Pv \in \operatorname{range} P$. Thus $V = \ker P + \operatorname{range} P$.
3. If $w \in \ker P \cap \operatorname{range} P$, write $w = Pu$. Then $w = Pu = P^2 u = Pw = 0$, so the sum is direct.

HW6 P2(a).

Idea. Differentiation lowers the degree of every nonconstant polynomial over $\mathbb{R}$ or $\mathbb{C}$.

1. If $Dp = \lambda p$ and $p \neq 0$ with $\lambda \neq 0$, a nonconstant p would give degree $\deg p - 1$ on the left and $\deg p$ on the right, a contradiction. A nonzero constant gives zero on the left and nonzero on the right, also impossible.
2. The eigenvalue 0 occurs because $D1 = 0$. By coefficient comparison, $Dp = 0$ iff p is constant, since the integers 1 through n are nonzero in either field.
3. Thus the only eigenvalue is 0 and $E(0, D) = \operatorname{span}(1)$; for $\lambda \neq 0$, $E(\lambda, D) = \{0\}$. This also covers $n = 0$.

HW6 P2(b).

Idea. Smooth functions include nonpolynomial exponential solutions of the derivative equation.

1. We cite elementary calculus: exponential differentiation, the product rule, and the fact that a function with derivative zero on $\mathbb{R}$ is constant. An eigenfunction is a nonzero function satisfying $Df = \lambda f$. On smooth real-valued functions, for every real λ the function $e^{\lambda x}$ is a nonzero eigenfunction because its derivative is $\lambda e^{\lambda x}$.
2. If $f' = \lambda f$, then $(e^{-\lambda x} f(x))' = 0$, so $f = ce^{\lambda x}$. Thus $E(\lambda, D) = \operatorname{span}(e^{\lambda x})$.
3. A complex-valued function is smooth when its real and imaginary parts are smooth; differentiation acts on those parts separately. For these functions, the same argument applies for every complex $\lambda = a + ib$, using $e^{\lambda x} = e^{ax}(\cos bx + i \sin bx)$ and the elementary trigonometric derivative rules. Hence all scalars of the selected field occur, unlike the polynomial case.

HW6 P3.

Idea. An invertible coordinate transformation transports each eigenspace.

1. $Rv = \lambda v$ iff $S^{-1}TSv = \lambda v$, iff $T(Sv) = \lambda Sv$ after multiplication by S.
2. Thus $v \in E(\lambda, R)$ iff $Sv \in E(\lambda, T)$, proving $E(\lambda, R) = S^{-1}E(\lambda, T)$.
3. Since S is invertible, it sends nonzero vectors to nonzero vectors and back. Therefore one eigenspace is nonzero iff the other is, giving equal eigenvalues.

HW6 P4.

Idea. For two-by-two matrices, noninvertibility is exactly a zero elementary determinant.

1. $tI - A = \left(\begin{smallmatrix} t-a & -b \\ -c & t-d \end{smallmatrix}\right)$, so its determinant is $(t-a)(t-d) - bc = t^2 - (a+d)t + (ad - bc)$.
2. λ is an eigenvalue iff $(\lambda I - A)v = 0$ for nonzero v, iff $\lambda I - A$ is not injective. Finite dimension makes this equivalent to noninvertibility.
3. Week 4's two-by-two criterion makes that equivalent to $\det(\lambda I - A) = 0$.

HW6 P5: A_1.

Idea. First specify the field, then solve the eigenvalue equation.

1. $\chi_{A_1}(t) = t^2 + 1$, so over $\mathbb{R}$ there are no eigenvalues and no eigenvector basis.
2. Over $\mathbb{C}$ the eigenvalues are $i, -i$. The equations $-y = \lambda x, x = \lambda y$ give $E(i, A_1) = \mathbb{C}(1, -i)$ and $E(-i, A_1) = \mathbb{C}(1, i)$.
3. The two eigenvalues are distinct, so their displayed eigenvectors are independent by Proposition 54 and form a basis of $\mathbb{C}^2$.

HW6 P5: A_2, A_3.

Idea. A root gives a candidate eigenvalue; the kernel dimension determines how many independent eigenvectors it supplies.

1. For A_2, $\chi(t) = (t-1)^2$ and $(A_2 - I)(x, y) = (y, 0)$. Thus the sole eigenvalue is 1, with $E(1, A_2) = \mathbb{R}e_1$. A one-dimensional eigenspace cannot supply a basis of $\mathbb{R}^2$.
2. For A_3, $\chi(t) = t^2 - t - 1$, with $\lambda_\pm = (1 \pm \sqrt{5})/2$. Solving gives $E(\lambda_\pm, A_3) = \mathbb{R}(1, \lambda_\pm)$.
3. These eigenvalues are distinct, so the two displayed eigenvectors form a basis of $\mathbb{R}^2$ by Proposition 54.

HW6 P6 (optional).

Idea. The first dependent iterate makes a finite initial span invariant.

1. A vector of $C(v, T)$ is a finite sum $\sum_{j=0}^{r} a_j T^j v$. Applying T gives $\sum_{j=0}^{r} a_j T^{j+1} v$, again a finite combination of the iterates. Thus the cyclic subspace is invariant.
2. Let $n = \dim V$. Since $v \neq 0$ and the first $n+1$ iterates are dependent, there is a first index m with $1 \leq m \leq n$ such that $T^m v \in \operatorname{span}(v, \ldots, T^{m-1} v)$.
3. The span U of those first m iterates is invariant: T sends each generator except the last to the next, and sends the last into U by the first-dependence relation.
4. Induction gives every later iterate in U, so $C(v, T) = U$, as required.

HW6 P7 (optional).

Idea. Stationarity forces equal coordinates, and total probability fixes their value.

1. Write $p = (a, b, c)^\mathsf{T}$. The first equation $a = (a+b)/2$ gives $a = b$; the third $c = (b+c)/2$ gives $c = b$.
2. The second equation $b = (a+c)/2$ then holds. Since $a + b + c = 1$, the probability vector is $p = (1/3, 1/3, 1/3)^\mathsf{T}$.
3. Thus if the chain starts uniformly over the three states, its distribution remains uniform after each transition.

Exam practice, Problem 7.

Idea. Invariance passes to annihilators because precomposition evaluates only on images that stay in the subspace.

1. If $v \in \ker T$, then $Tv = 0 \in \ker T$; if $w = Tv \in \operatorname{range} T$, then $Tw = T(Tv) \in \operatorname{range} T$. Thus both subspaces are invariant.
2. More generally, let U be invariant and $\varphi \in U^0$. For each $u \in U$, $Tu \in U$, so $(T'\varphi)(u) = \varphi(Tu) = 0$.
3. Therefore $T'\varphi \in U^0$, proving annihilator invariance. Apply this to $U = \ker T$ and $U = \operatorname{range} T$.